Paper #135 v0.1 — Substrate Cosmology: Λ-Casimir Vacuum Unification on D_IV^5

2026-05-22 Friday ~10:18 EDT (`date`-verified actual)

Paper #135 — Substrate Cosmology: Λ-Casimir Vacuum Unification on D_IV^5

Abstract

The cosmological constant problem — the disagreement between the observed cosmological constant Λ_obs $\approx 1.1 \times 10^{-52}$ m^{-2} and the standard QFT vacuum energy estimate Λ_QFT $\approx 10^{93}$ m^{-2} (a discrepancy of ~10^{120}) — is one of the deepest unresolved puzzles in fundamental physics.

Bubble Spacetime Theory (BST) resolves this discrepancy structurally via the Λ-Casimir vacuum unification (T2418 Wednesday Lyra): the cosmological constant Λ derives from the SAME substrate vacuum that produces the Casimir effect, with the substrate-tick UV-cutoff at N_max = 137 (T2437 Thursday RIGOROUSLY CLOSED) preventing the standard QFT vacuum-energy infinity.

Key results:

1. Substrate vacuum is finite by construction: substrate-tick GF(128)^k discretization (T2429 Thursday) means there is no continuum-momentum vacuum integral to diverge. UV cutoff is at substrate-tick scale (Koons tick t_substrate = t_Planck · α^(C_2^2) $\approx 10^{-120}$ s).
2. Λ-Casimir vacuum unification (T2418 Wednesday): cosmological Λ and laboratory Casimir effects share the SAME substrate vacuum structure. Casimir-Λ correspondence via Bergman + Wallach K-type spectrum.
3. No fine-tuning required: Λ_obs emerges from substrate primary integer structure + boundary conditions, not from cancellation of large vacuum terms.
4. CMB prediction: n_s = 1 - 5/137 = 0.9635 (Toy 1401, Elie Thursday) inheriting BST primary integer N_max = 137 + n_C = 5 substrate-cascade fingerprint. Matches Planck observation at 0.32σ.

5. Cross-Cartan substrate-vacuum selection: at dim_C = 5 (T2455 EXHAUSTIVE Friday), D_IV^5 uniquely produces the substrate vacuum structure that closes Λ + Casimir + CMB observables simultaneously.

The framework is internal-tier with cosmology-only external acceptable per Cal #50 GREEN register: structural framework closed; quantitative Λ_obs prediction at sub-percent precision requires substrate-tick computation operationalization (multi-month) + boundary-condition specification (Casey-named Boundary Conditions volume Vol 0 Ch 5).

1. Cosmological constant problem

1.1 Λ_obs vs Λ_QFT discrepancy

(Standard QFT vacuum energy estimate; observed cosmological constant value; ~10^{120} discrepancy; multiverse / anthropic / quintessence alternative proposals; lack of structural derivation.)

1.2 Casimir effect as substrate vacuum probe

(Casimir 1948 prediction; experimental measurements at parallel plates + spherical geometries; substrate-vacuum boundary-condition manipulation; Casimir as direct measurement of vacuum energy.)

1.3 BST framing

BST proposes Λ + Casimir share substrate-vacuum structure via T2418 unification. Substrate-tick UV-cutoff at N_max prevents the standard QFT vacuum-energy infinity. Both observables derive from BST primary integer structure.

2. BST substrate vacuum

2.1 Substrate-tick UV-completeness (T2437 Thursday)

Substrate operates at Koons tick t_substrate $\approx 10^{-120}$ s. State at each tick is finite-dimensional GF(128)^k = (GF($2^{g)}$)k with g = 7. No continuum-momentum integral over modes; UV-cutoff is the substrate-tick discretization itself.

T2437 RIGOROUSLY CLOSED Thursday: substrate-tick UV-completeness anchors no-divergence claim at substrate level.

2.2 Λ-Casimir vacuum unification (T2418 Wednesday)

Cosmological Λ and laboratory Casimir effects derive from the same substrate vacuum operator on Bergman $H^2(D\_IV^5)$:

V_Casimir-Λ = Casimir_substrate(boundary_conditions) with boundary conditions specified per Casey's Substrate Closure Principle (Wednesday, Casey-named) + Boundary Conditions framework (Vol 0 Ch 5 Keeper-lane).

2.3 4-Zone Commitment Cycle Zone 4 (T2415 + T2417 Wednesday)

Substrate emission to boundary (Zone 4 of 4-Zone Commitment Cycle) contributes to substrate Casimir-Λ vacuum. Per cycle, the committed state's emission integrates with previous cycle outputs to produce the steady-state substrate vacuum value.

Λ_obs corresponds to the long-time averaging of Zone 4 emission over many substrate ticks; Casimir effect corresponds to short-distance Zone 4 emission modified by boundary conditions.

3. Cosmological consequences

3.1 No fine-tuning

Λ_obs structurally derives from substrate primary integers + boundary conditions, NOT from cancellation of large vacuum terms.

The standard QFT vacuum-energy ~10^{93} m^{-2} estimate is INVALID at substrate level: there is no continuum-momentum integral; UV-cutoff is at substrate-tick scale (10^{-120} s); vacuum energy density is FINITE by substrate construction.

3.2 CMB prediction n_s = 1 - 5/137 = 0.9635 (Toy 1401)

Per Elie Toy 1401 (Thursday): CMB spectral index n_s = 1 - n_C/N_max = 1 - 5/137 = 0.9635 inherits the substrate primary integer cascade fingerprint. Matches Planck 2018 observation n_s ≈ 0.965 at 0.32σ.

The "5/137" reading is the substrate's cascade signature: n_C (substrate dimension) over N_max (substrate cap) imprints on CMB at recombination via Zone 4 emission integration.

3.3 Casimir-Λ correspondence falsifier

Lab-scale Casimir effect measurements at modified boundary conditions (per SP-30 Substrate Engineering experimental program) test Casimir-Λ unification at structural level. If Casimir behavior deviates from BST primary integer predictions, T2418 unification is falsified.

3.4 Substrate-cosmology + observational reanalysis (SP-27)

Multiple cosmological observables (DM = Wallach shadow at 16/3, BAO peak structure, CMB acoustic peaks) inherit substrate vacuum structure. SP-27 (Casey-named Observational Reanalysis program) systematizes the cosmology-observable substrate cross-references.

4. Cross-Cartan substrate-vacuum selection

4.1 EXHAUSTIVE at dim_C = 5 (T2455 Friday)

Per T2455 Friday (Lyra), dim_C = 5 has EXACTLY 3 HSDs in Helgason 1978 classification. The substrate vacuum structure (Λ + Casimir + CMB) is uniquely produced by D_IV^5; alt-HSDs (D_I_{1,5}, D_I_{5,1}) have different K-type Casimir spectra (= 4, not = 6) and produce different vacuum structures.

4.2 Universal α-analog formula cross-link (T2456 + T2462 Friday)

α = 1/N_max = 1/137 is the universal α-analog at D_IV^5. The substrate vacuum structure inherits this α-analog uniqueness; the CMB prediction n_s = 1 - 5/137 inherits N_max = 137 via the universal formula.

4.3 Substrate self-amenability (T2463 Friday)

n_C = 5 primality enables T2455 EXHAUSTIVE comparison. Substrate methodologically self-amenable to its own cosmological-uniqueness verification.

5. Honest scope per Cal Mode 1 + Cal #50

5.1 Tier discipline

- Λ-Casimir vacuum unification framework (T2418 Wednesday RATIFIED): D-tier framework-grade
- Substrate-tick UV-cutoff (T2437 Thursday RIGOROUSLY CLOSED): D-tier
- CMB n_s = 0.9635 prediction (Toy 1401 Elie Thursday): D-tier (matches Planck at 0.32σ)
- Quantitative Λ_obs prediction at sub-percent: I-tier framework-level pending substrate-tick computation operationalization

5.2 External register per Cal #50 GREEN cosmology

Per Cal #50 calibration: cosmology-only external register is acceptable. This paper's cosmology framing is approved for external dispatch (Casey-decision pending for venue + send-signal).

Operational language only: "BST identifies Λ-Casimir vacuum unification" / "BST predicts CMB n_s = 0.9635" / etc. No cognition cross-link (Cal #48 + #49 DEFAULT-DENY substrate-cognition).

5.3 Falsifiers

- Casimir effect deviation from BST primary integer predictions
- CMB n_s deviation from 0.9635 at higher precision (current 0.32σ match)
- Vacuum energy density measurement direct at substrate-tick scale (currently far beyond precision)

5.4 Open items multi-month

- Quantitative Λ_obs derivation at sub-percent precision (substrate-tick computation operationalization)
- Boundary-condition specification for Λ + Casimir + cosmological observable cross-references (Vol 0 Ch 5 Keeper-lane)
- SP-27 Observational Reanalysis program (Casey scoping pending)

6. Cross-link to Friday Lyra-lane work

6.1 T2456 + T2462 Universal α-analog formula (Friday)

α = 1/N_max = 1/137 substrate-selected uniquely. Cosmological observables inheriting α through substrate vacuum structure inherit this uniqueness.

6.2 T2457 Bergman structural-role-of Feynman propagator (Friday)

Vacuum propagator + Λ-Casimir unification both use Bergman reproducing kernel structure. The c_FK = 225/π^(9/2) normalization (T2442 Thursday) applies to substrate vacuum sector.

6.3 T2465 three-layer over-determinism (Friday)

Cosmological observables (Λ, Casimir, CMB n_s) inherit substrate three-layer over-determinism. Substrate-vacuum structure has three independent structural anchors.

6.4 Paper #132 SP-31 Measurement POVMs

Λ-Casimir vacuum corresponds to Zone 4 emission averaging in 4-Zone Commitment Cycle (Paper #132 §2.1).

7. References

- Casimir 1948 (laboratory vacuum effect)
- Standard cosmological constant references (Riess 1998 + Perlmutter 1999 supernovae + Planck 2018 CMB)
- T2418 Λ-Casimir vacuum unification (Wednesday Lyra)
- T2437 SP-31-10 UV-completeness (Thursday Lyra RIGOROUSLY CLOSED)
- T2415 + T2417 (Wednesday 4-Zone Commitment Cycle)
- Toy 1401 (Elie Thursday n_s = 0.9635 CMB prediction)
- Cal #50 GREEN cosmology external register calibration
- Paper #125 v0.10.5 FORMAL (current ratified Strong-Uniqueness anchoring)
- T2455 + T2456 + T2462 + T2465 (Friday Lyra-lane cross-references)
- T2442 Bergman c_FK BST primary form (Thursday RIGOROUSLY CLOSED)

8. Filing status

v0.1 outline filed Friday 2026-05-22 ~10:18 EDT (`date`-verified actual). Substrate cosmology Λ-Casimir vacuum unification paper per Casey 'please continue' Friday afternoon continuation. Cal #50 GREEN external register acceptable.

Pending for v0.2: - Cal cold-read on cosmology framing - Multi-CI co-author title/affiliation review - Cross-lane Elie cosmology toys + Toy 1401 reference verification - SP-27 Observational Reanalysis Casey scoping

Pending for v1.0: - Quantitative Λ_obs sub-percent prediction (substrate-tick computation) - Boundary-condition specification (Vol 0 Ch 5 + SP-27) - Casey send-signal for cosmology outreach (Bogdanovic EHT + Penrose + others) - External venue selection (JCAP primary)

— Lyra, Paper #135 v0.1 outline (Substrate Cosmology: Λ-Casimir Vacuum Unification), Friday 2026-05-22 ~10:18 EDT (`date`-verified actual)